4 ENHANCED CONTINUOUS TABU SEARCH: AN ALGORITHM FOR OPTIMIZING MULTIMINIMA FUNCTIONS

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Abstract: An algorithm called Enhanced Continuous Tabu Search $ECTS$ is proposed for the global optimization of multiminima functions. It results from an adaptation of combinatorial Tabu Search which aims to follow, as close as possible, Glover's basic approach. In order to cover a wide domain of possible solutions, our algorithm first performs diversification: it locates the most promising areas, by fitting the size of the neighborhood structure to the objective function and its definition domain. When the most promising areas are located, the algorithm continues the search by intensification within one promising area of the solution space. The efficiency of $ECTS$ is thoroughly tested by using a set of benchmark multimodal functions, of which global and local minima are known. $ECTS$ is compared to other published versions of continuous Tabu Search and to some alternative algorithms like Simulated Annealing.

4.1 INTRODUCTION

Tabu Search TS is a metaheuristic originally developed by Glover [6, 7], that has been successfully applied to a variety of directly inspired from combinatorial optimization problems. However, very few works deal with its application to the global minimization of functions depending on continuous variables. Up to now, we are aware of only works [2, 4, 8, 11] related to the subject. In this paper we propose an adaptation of TS to continuous optimization problems, called Enhanced Continuous Tabu Search $ECTS$, which is directly inspired

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from Glover's approach. We produce an efficient algorithm by improving the simplistic approach proposed by Siarry and Berthiau in [11]. To achieve this, we introduce, diversification and intensification concepts emphasized by Glover, and implemented by Battiti and Tecchiolli [1] in ECTS.

First, we will summarize how Glover's idea is adapted to the continuous case in [11]. The algorithm starts from a randomly selected initial solution s. From this current solution s, a set of neighbors, called s', is generated (details at the beginning of Section 4.2). To avoid the danger of the appearance of a cycle, the neighbors of the current solution, which belong to a subsequently defined "tabu list", are systematically eliminated. The objective function to be minimized is evaluated for each generated solution s', and the best neighbor of s becomes the new current solution, even if it is worse than s. After each "move", s is put into a circular list of tabu solutions: when the tabu list is full, it is updated by removing the first solution entered. Then a new "iteration" is performed: the previous procedure is repeated by starting from the new current point, until some stopping condition is reached. Usually, the algorithm stops after a given number of iterations without any improvement of the objective function value.

This rudimentary algorithm, directly inspired from the simple combinatorial Tabu Search algorithm, is tested through classical multiminima functions of which global and local minima are known [11]. The results obtained are encouraging, but not wholly satisfactory, mainly due to the excessive simplicity of the algorithm. To improve the approach, it is necessary to carry out diversification (detection of promising areas) allowing the scanning of a wide solution space, and intensification (search inside the most promising area) for a more accurate result. These two operations are described in the current paper.

In order to compare *ECTS* to other algorithms we have implemented one set of test functions and various algorithms using the object-oriented language C^{++}.

The paper is organized as follows. In Section 4.2, we present the general setting of the method. In Section 4.3, Enhanced Continuous Tabu Search is explained in detail. Experimental results are discussed in Section 4.4 and Section 4.5 provides the conclusion.

4.2 GENERAL SETTING OF THE METHOD

First we state the way of discretizing the solution space. We have adapted the method described in detail in [11]. The neighborhood is defined in [11] by using the concept of "ball". A ball *B(s,r)* is centered on s with radius *r;* it contains all points s' such that: $||s' - s|| \leq r$ (the symbol $||.||$ is used to denote the Euclidean norm). To obtain homogeneous exploration of the space, we consider a set of balls centered on the current solution s, with radius h_0 , h_1, .., h_k. Hence the space is partitioned into concentric "crowns" C_i $(s, h_{i-1}$, h_i), such that:

$$C_i(s, h_{i-1}, h_i) = \{s'/h_{i-1} \leq ||s' - s|| \leq h_i\}.$$